Xiaohan Zhang

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EDUCATION BACKGROUND

Ph.D. in Computer Science, SUNY Binghamton, Binghamton, NY *Aug. 2020 – Present*

Advisor: Prof. Shiqi Zhang

M.S. in Computer Science, SUNY Binghamton, Binghamton, NY *Aug. 2019 – May 2020*

B.S. in Information Security, Renmin University of China, Beijing, China *Sep. 2015 – May 2019*

RESEARCH INTERESTS

Robotics and Artificial Intelligence

In particular, I am interested in developing algorithms that integrate planning (probabilistic planning, task-motion planning) and learning (classical machine learning, deep learning, reinforcement learning) for robot perception.

PUBLICATIONS

Peer-Reviewed Conference Papers:

- Kishan Chandan, Jack Albertson, **Xiaohan Zhang**, Xiaoyang Zhang, Yao Liu, and Shiqi Zhang, “Learning to Guide Human Attention on Mobile Telepresence Robots with 360 Vision.” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021
- **Xiaohan Zhang**, Jivko Sinapov, and Shiqi Zhang, “Planning Multimodal Exploratory Actions for Online Robot Attribute Learning.” Robotics: Science and Systems (RSS), 2021
- Yan Ding, **Xiaohan Zhang**, Xingyue Zhan, and Shiqi Zhang, “Task-Motion Planning for Safe and Efficient Urban Driving.” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020

Peer-Reviewed Workshop Papers:

- **Xiaohan Zhang**, Jivko Sinapov, and Shiqi Zhang, “Information-Theoretic Reward Shaping for Multimodal Object Attribute Learning.” Workshop on Integrating Planning and Learning, Robotics: Science and Systems (RSS Workshop), 2021
- Kishan Chandan, **Xiaohan Zhang**, Jack Albertson, Xiaoyang Zhang, Yao Liu, and Shiqi Zhang, “Guided 360-Degree Visual Perception for Mobile Telepresence Robots.” Workshop on Closing the Academia to Real-World Gap in Service Robotics, Robotics: Science and Systems (RSS Workshop), 2020

Preprints & Technical Reports:

- Saeid Amiri, Yan Ding, **Xiaohan Zhang**, Hao Yang, Chad Esselink, and Shiqi Zhang, "Open-World Robot Task Planning with Domain and Commonsense Knowledge." Submitted to International Conference on Principles of Knowledge Representation and Reasoning (KR), 2021
- **Xiaohan Zhang** and Jinchuan Chen, "Blockchain-based Data Protection System for the Internet of Things." Bachelor's Thesis, Renmin University of China, 2019

PROFESSIONAL EXPERIENCE

Graduate Research Assistant Binghamton, NY
Autonomous Intelligent Robotics Laboratory, SUNY Binghamton *Jan. 2021 – Present*

- Learning to Perceive for Efficient Task and Motion Planning
  - Proposed a task-level-optimal framework for learning the state-mapping function using predicted heatmaps from a Fully Convolutional Network (FCN)
  - Demonstrated the proposed framework on a Segway-based mobile robot equipped with a UR5e arm in Gazebo
- Robot Attribute Learning and Identification
  - Proposed a POMDP-based information-theoretic reward shaping algorithm that equips a robot with the capability of co-optimizing its sequential action selection toward (efficiently and accurately) learning and identifying object attributes simultaneously

Graduate Teaching Assistant Binghamton, NY
SUNY Binghamton *Jan. 2020 – Dec. 2020*

- CS 465/565 - Intro to Artificial Intelligence

Software Engineer Intern Beijing, China
BroadStar Information Technology Co., Ltd *May 2017 – Sep. 2017*

- Designed an authentication key management system based on BroadStar COS operating system

Course Assistant Beijing, China
Renmin University of China *Sep. 2016 – Jan. 2017*

- Probability and Mathematical Statistics

TECHNICAL SKILLS

- **Language:** C/C++, Python, Java, JavaScript, HTML, PHP, SQL, Solidity
- **Software and Tool:** ROS, CARLA, Gazebo, Unity, Clingo, Git, Vim, LaTeX